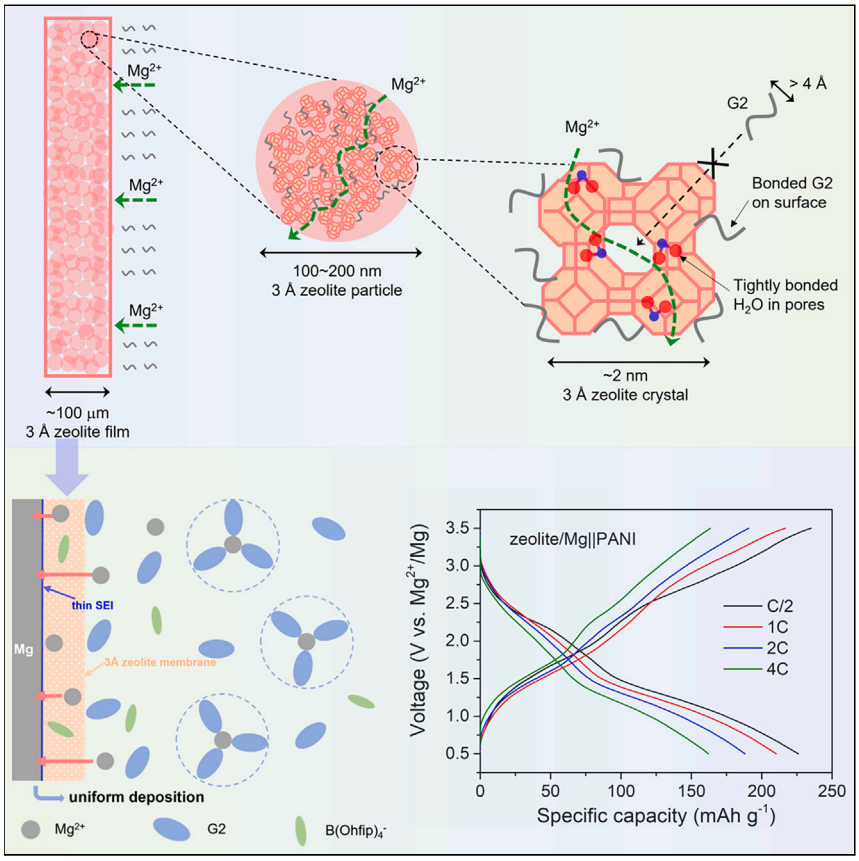
Joule

CellPress

Article

Stabilizing magnesium plating by a low-cost inorganic surface membrane for high-voltage and high-power Mg batteries



Chloride-free Mg electrolytes are critical to achieve high-voltage Mg batteries without corrosion, but the short lifetime of Mg metal anodes induced by electrolyte decomposition presents a serious challenge. This work reports on a practical Mg^{2+}-ion conducting and cost-effective protective interface comprising a zeolite/polymer membrane that traps free solvent and thus limits its decomposition. Moreover, it assists the facile transport of Mg^{2+} cations, enabling fast, long-term plating/stripping of Mg through the interface, and supports full battery cycling at fast rates.

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Highlights

A zeolite 3A/polymer membrane functions as a protective interface for Mg anodes

Decomposition of free diglyme is suppressed at the Mg/electrolyte interface

Highly reversible Mg plating/stripping through the membrane over 6,000 h

Full cells show stable cycling at a 3.5 V cutoff voltage at high rates (4C)

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